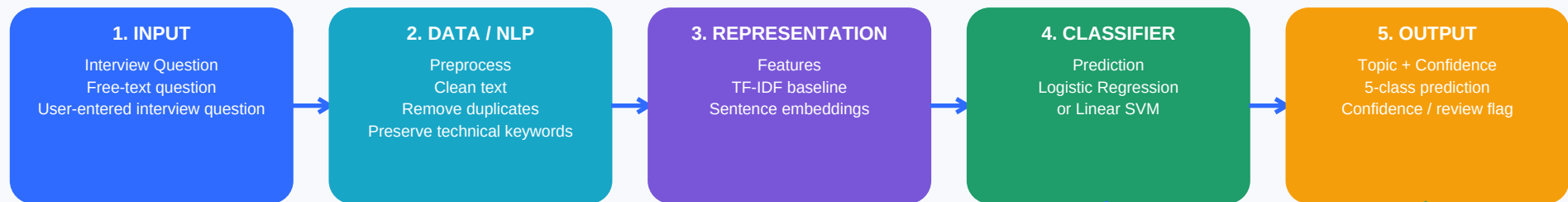
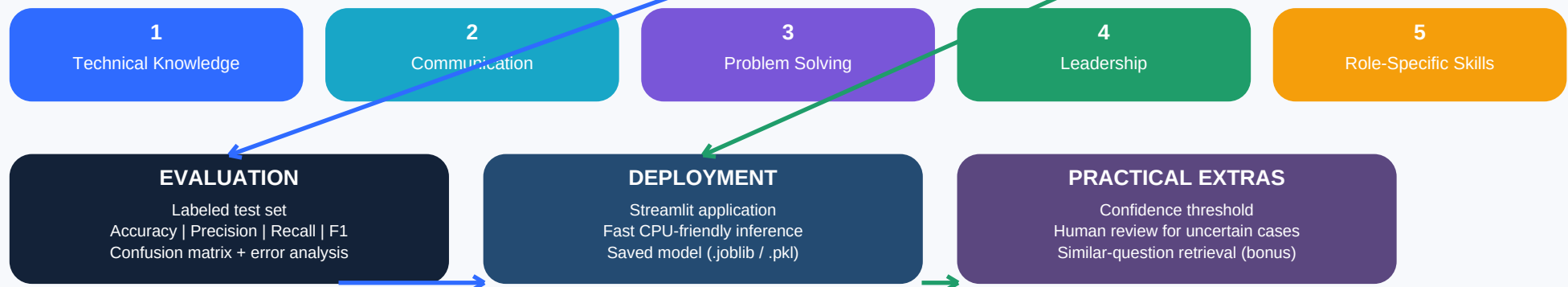


EMP-26 Specification Flowchart

End-to-end pipeline for a lightweight interview-question topic classifier



Required classification categories



Recommended Project Structure

Organize data, training, evaluation, models, and deployment as separate modules

```
EMP-26/
+-- data/
|   +-- raw/
|   |   -- raw_questions.csv
|   +-- processed/
|   |   -- labeled_questions.csv
|   +-- splits/
|   |   +-- train.csv
|   |   +-- validation.csv
|   |   -- test.csv
+-- notebooks/
|   +-- 01_data_exploration.ipynb
|   +-- 02_preprocessing.ipynb
|   +-- 03_baseline_model.ipynb
|   +-- 04_embedding_model.ipynb
|   -- 05_evaluation.ipynb
+-- src/
|   +-- preprocessing.py
|   +-- train.py
|   +-- predict.py
|   +-- evaluate.py
|   -- utils.py
+-- models/
|   +-- tfidf_model.joblib
|   -- embedding_classifier.joblib
+-- app/
|   -- app.py
+-- reports/
|   +-- confusion_matrix.png
|   -- model_comparison.png
+-- requirements.txt
-- README.md
```

A. DATA LAYER

- Collect questions
- Label with 5 classes
- Clean + deduplicate
- Stratified train/val/test split

B. ML LAYER

- Baseline: TF-IDF + Logistic Regression
- Final: Embedding + Logistic Regression / SVM
- Tune and save best model



C. EVALUATION LAYER

- Accuracy, precision, recall, F1
- Class-wise results
- Confusion matrix
- Error analysis



D. APPLICATION LAYER

- Streamlit UI
- Single-question prediction
- Confidence scores
- Low-confidence human review



Specification Requirements Map

FUNCTIONAL REQUIREMENTS

- Input: text interview question
- Classify into exactly one of 5 topics
- Return predicted label
- Optionally return confidence distribution
- Low-confidence cases can be flagged for review

DATASET REQUIREMENTS

- CSV with question + label fields
- Labels mapped to 5 required classes
- Remove duplicates / missing rows
- Keep class counts reasonably balanced
- Target: 3,000-5,000+ labeled questions

MODEL REQUIREMENTS

- Lightweight and CPU-friendly
- Baseline: TF-IDF + Logistic Regression
- Final candidate: embeddings + linear classifier
- Reproducible training pipeline
- Save trained model for inference

EVALUATION REQUIREMENTS

- Use held-out labeled test data
- Accuracy
- Precision, Recall, F1-score
- Confusion matrix
- Class-wise error analysis

SOFTWARE / HARDWARE

- Python 3.10+
- pandas, numpy, scikit-learn
- sentence-transformers (for semantic model)
- Streamlit for UI
- 8 GB RAM minimum; GPU optional

DELIVERABLES

- Labeled dataset
- Training + evaluation code
- Saved model artifacts
- Streamlit demo
- Charts / metrics / README / presentation

MVP

3,000-5,000+ labels | 70/15/15 split | Baseline + final model | Metrics + confusion matrix | Streamlit demo